

PROFESSIONAL SUMMARY

- Cloud and backend platform engineer with 5+ years designing, deploying, and operating production cloud infrastructure, Python/FastAPI services, and secure APIs across AWS and Azure — with directly transferable containerisation, IAM, and observability experience to GCP-native environments.
- Hands-on authentication and identity engineering: implemented OAuth 2.0 / JWT, RBAC, Azure AD / SSO-compatible auth flows, and service-level security patterns across production multi-service platforms.
- Deep Python backend expertise: async FastAPI services, OpenAPI documentation, SQL bulk operations, and 97% database performance optimisation through asyncio refactoring — the exact async patterns and API standards this role requires.
- Embedded in the ASU ecosystem as an MS alumnus, HCI researcher, and Engineering Tutor — committed to the mission of access and excellence that drives both ASU and ASU Prep.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, S3, IAM, VPC, CloudWatch), Azure (VM, Service Bus, Blob Storage, Azure AD) — GCP-transferable via Docker / Kubernetes / containerised service deployment

Identity & Auth: OAuth 2.0, JWT, RBAC, Azure AD / Active Directory, SSO, service account patterns, secure API authentication and authorisation

Backend & APIs: Python (FastAPI, asyncio), REST APIs (OpenAPI/Swagger), GraphQL, Node.js, Express.js — API versioning, documentation, caching, pagination, performance tuning

Observability: EFK Stack (Elasticsearch, Fluentd, Kibana) — centralised logging, distributed tracing, metrics, alerting, real-time pipeline monitoring

Infrastructure & DevOps: Docker, Kubernetes, GitHub Actions (CI/CD), Linux server management, Bash / Python scripting, TLS / secure hosting

Data & Databases: PostgreSQL (pgvector), MySQL, MongoDB, SQL — data modeling, query optimisation, ORM, BigQuery-compatible SQL patterns

Security: IAM principles, RBAC, secrets management, secure service-to-service communication, web security best practices

EXPERIENCE

HCI Researcher | Jan 2026 – Present | Arizona State University | Tempe, AZ

Active researcher within the ASU ecosystem — building and evaluating AI-integrated applications with a focus on trustable, accessible user experiences.

- Developing an Android application that generates context-aware LLM responses — architecting secure API integrations and managing service authentication from client to backend.

Lead Developer | Aug 2025 – Dec 2025 | ABC Academy of Music | MS Industry Capstone | Phoenix, AZ

Led a team of 5 engineers designing and operating a production cloud-hosted web platform — owning infrastructure, CI/CD, database architecture, and stakeholder communication.

- Designed a PostgreSQL relational database schema and Java Spring Boot REST API backend — defining data models, access patterns, and API contracts for a production multi-user application.
- Deployed and operated the platform on AWS EC2 Linux servers — configuring CI/CD pipelines via GitHub Actions, managing server processes, and achieving 99% platform availability with monitoring and alerting in place.
- Served as the primary technical liaison to the product manager — translating requirements into architecture decisions and communicating trade-offs clearly across a cross-functional team.

Lead Developer | Apr 2023 – Dec 2023 | Tecnotree | Bengaluru, KA

Awarded "Spotlight Top Performer" for leading high-quality, reliable platform feature delivery with minimal post-release defects.

Senior Software Engineer | Jul 2022 – Mar 2023 | PangeaTech | Bengaluru, KA

Founding engineer on a production ML SaaS platform — owned full cloud platform architecture, Python backend services, identity/auth layer, and observability infrastructure.

- **Cloud platform architecture:** Designed and operated Azure-hosted services including VM infrastructure, Azure Service Bus for event-driven messaging, and Blob Storage — managing networking, scaling, and deployment across all platform services.
- **Identity and auth:** Implemented OAuth 2.0 / JWT-based authentication securing all REST API endpoints, with Azure AD integration and role-based access control enforced across the platform — directly applicable to federated identity and SSO work.
- **Real-time observability:** Deployed a centralised EFK stack (Elasticsearch, Fluentd, Kibana) on Linux — establishing logging, distributed tracing, and alerting pipelines that reduced incident root-cause time from 4 hours to real time.
- **30% cloud cost reduction:** Optimised VM right-sizing and process scaling across Azure infrastructure — demonstrating the cost-conscious cloud operations mindset this role requires.
- Architected event-driven Python data pipelines using Azure Service Bus for async workload distribution across Kubernetes nodes — reducing data processing time by 80% and demonstrating async platform engineering at scale.

- Authored 50+ code reviews and mentored engineers on backend API design, cloud security patterns, and Python best practices.

Software Engineer | Nov 2021 – Jul 2022 | PangeaTech | Bengaluru, KA

Primary Python backend and infrastructure engineer — owned database systems, async service design, containerisation, and CI/CD operations.

- **97% Python async performance gain:** Refactored synchronous Python ETL services to fully asynchronous Python (asyncio) — reducing database refresh time from 8 hours to 15 minutes, the exact async FastAPI pattern central to this role.
- Designed and maintained MySQL stored procedures and 50+ SQL materialized views for high-throughput data access — applying query optimisation, bulk insert patterns, and data integrity controls at production scale.
- **Maintained servers at 99% availability** on Linux — managing database processes, writing Bash and Python automation scripts, monitoring service health, and resolving production incidents.
- **85% faster deployments:** Containerised microservices with Docker, orchestrated with Kubernetes, and automated CI/CD via GitHub Actions — the same containerisation stack that maps directly to Cloud Run / GKE on GCP.

Software Engineer | Jul 2019 – Apr 2021 | Tenable (a JPMorgan Chase company) | Pune, MH

- Migrated a microservice from REST to GraphQL, reducing network overhead by 35% — demonstrating API design judgment and a focus on backend performance optimisation.
- Maintained security and compliance standards across a high-traffic platform handling sensitive customer data for millions of users — applying secure development practices throughout the full stack.

PROJECTS & ADDITIONAL EXPERIENCE

Compulator | compulator.app | Live cloud-hosted platform

- Sole owner of cloud infrastructure for a live public-facing platform — managing Azure VM deployment, CI/CD pipelines, TLS/secure HTTPS hosting, and site reliability. Demonstrates end-to-end ownership of cloud platform operations.

Travel Recommendations MCP | TypeScript REST API service

- Built a fault-tolerant API service with rate limiting, fallback handling, and structured error responses — designing for reliability and resilience in a multi-consumer service architecture.

Engineering Tutor | ASU: Delivered 100+ sessions on Databases, Distributed Systems, Operating Systems, and Data Structures — deepening system-level expertise and contributing to the ASU academic community.

CERTIFICATIONS

Foundations of AI Engineering | CodePath | Spring '26 • Applications of AI Engineering | CodePath | Summer '26

EDUCATION

Master of Science in Software Engineering, GPA: 3.87 / 4.0 | Dec 2025 | Arizona State University | Mesa, AZ

Coursework: System Design, Distributed Software Systems, Data Structures and Algorithms, Agile Software Development

Bachelor of Science in Computer Science | Aug 2019 | REVA University | Bengaluru, KA